

Acme Corp - Nexus Integration Hub

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Data Model & Schema Mapping

Overview

The Nexus Integration Hub maps legacy mainframe data structures and SaaS API payloads to Acme Corp canonical v2 schema. This document describes mapping rules and validation logic.

Source Systems

SAP ERP — Customer master, order headers, inventory (IDoc format)

Salesforce CRM — Accounts, opportunities, contacts (REST API v58)

Workday HRIS — Employee records, org units (SOAP API)

Legacy Mainframe — Fixed-width EBCDIC tables (CUSTMAST, ORDHDR, ORDDTL, INVMST)

Target Schema (Canonical v2)

Modern tables use UTF-8, JSONB metadata columns, and UUID primary keys.

customers from SAP KNA1 and legacy CUSTMAST

orders from SAP VBAK plus ORDDTL line items nested as JSONB

employees from Workday Worker API

Validation Rules

Before a sync job runs, the engine validates:

1. Schema compatibility — all source fields have a mapping
2. Referential integrity — foreign keys resolvable in target
3. Data type coercion — no silent truncation
4. Duplicate detection — primary key uniqueness in source batch

Rollback Snapshots

Point-in-time snapshots stored in S3 as Parquet files, 30-day retention, monthly restore tests.

Performance Tuning

batch_size default 1000, recommended 5000 for large jobs

worker_count default 4, recommended 8

parallel_entities default 1, recommended 3 for jobs over 1M records